

DBMS Lecture 11: Foreign Key & Referential Integrity

Deep-Dive Notes on DML Operations, Cascading Actions & Exam Patterns

1 1. Introduction & Core Significance

While Lecture 10 introduced the structural concept of a Foreign Key, this lecture focuses on its primary operational purpose: **maintaining Referential Integrity** during data modification operations (INSERT, DELETE, and UPDATE).

Referential Integrity mechanics are extensively evaluated in:

- **GATE, UGC NET, ISRO, and PSU Exams** (Operation risk evaluation questions)
- **SQL DML Execution Analysis** (CASCADE, SET NULL, RESTRICT)
- **Database System Design** (Preventing dangling references and orphan rows)
- **Core Technical Interviews** (Predicting database behavior during operations)

2 2. What is Referential Integrity?

Referential Integrity Definition

Referential Integrity is a relational database rule requiring that every foreign key value in a referencing relation must either:

- i. Point to a valid, existing candidate/primary key value in the referenced relation, **OR**
- ii. Be completely NULL (if the foreign key attribute permits nulls).

This ensures that relationships between tables remain consistent and prevents *dangling pointers* or orphan child records.

Integrity Constraint Distinction

Primary Key	⇒	Entity Integrity Rule (No NULLs, Unique Identity)
Foreign Key	⇒	Referential Integrity Rule (Valid Cross-Table References)

3 3. Motivational Example: University Schema

Consider a database with a parent table `Student` and a child table `Course`:

Parent / Referenced Table: `Student`

Roll_No (PK)	Name	Address
1	A	Delhi
2	B	Mumbai
3	C	Pune

Child / Referencing Table: Course

Course_ID (PK)	Course_Name	Roll_No (FK)
C1	DBMS	1
C2	Networks	2
C3	Java	1

4 4. Structural Relationship & Column Naming Flexibility

In this architecture:

- Student is the **Referenced (Parent) Table** because it owns the primary key (Roll_No).
- Course is the **Referencing (Child) Table** because its Roll_No attribute targets the parent table.

Important Note on Attribute Names

The foreign key column in the child table **does not** need to share the exact same column name as the referenced parent key. For example, Course.Student_ID (FK) referencing Student.Roll_No (PK) is completely valid as long as their data types and domains match.

5 5. Operational Analysis Matrix: INSERT, DELETE, UPDATE

When executing SQL DML operations, the DBMS checks referential integrity rules. The key question for exams is: **Which operations on which table risk causing a violation?**

Operation	Referenced (Parent) Table	Referencing (Child) Table
INSERT	✓ Safe (No Risk)	Violation Risk
DELETE	Violation Risk	✓ Safe (No Risk)
UPDATE	Violation Risk	Violation Risk

6 6. Operations on Referenced (Parent) Table

6.1 INSERT in Parent Table (✓ Safe)

- **Scenario:** Inserting a new student record into Student (Roll_No = 4, Name = 'D').
- **Result:** Allowed without violation.
- **Reason:** The parent record has no dependency on child records. A student can exist in the system even if they are not currently enrolled in any courses.

6.2 DELETE from Parent Table (Violation Risk)

- **Scenario A (Referenced Key Deletion):** Deleting Student where Roll_No = 1.
 - Course contains rows (C1 and C3) referencing Roll_No = 1.
 - Deleting student 1 would leave C1 and C3 referencing a nonexistent parent (*orphan rows*).
 - **Result:** Causes a Referential Integrity Violation (unless a referential action handles it).
- **Scenario B (Unreferenced Key Deletion):** Deleting Student where Roll_No = 3.

- No child rows in Course reference Roll_No = 3.
- **Result: Allowed safely.**

6.3 UPDATE in Parent Table (Violation Risk)

- **Scenario:** Updating Student.Roll_No from 2 to 20.
- **Result: Causes a Violation** because Course row C2 references Roll_No = 2, which no longer exists after the update.

7 7. Handling Parent Deletions & Updates (Referential Actions)

To handle potential violations caused by parent deletions or updates, SQL allows configuring explicit **Referential Actions**:

ON DELETE / UPDATE RESTRICT or NO ACTION (Default)

The DBMS strictly rejects the deletion/update on the parent table if any child row references that key.

ON DELETE CASCADE

Automatically deletes all child rows referencing the deleted parent key.

Delete Student 1 $\Rightarrow$ DBMS automatically deletes Course C1 and Course C3

ON DELETE SET NULL

Automatically sets the foreign key value in dependent child rows to NULL when the parent row is deleted.

Constraint Requirement for SET NULL

SET NULL requires that the foreign key attribute in the child table allows NULL values (i.e., not constrained by NOT NULL or part of a composite primary key).

ON DELETE SET DEFAULT

Sets the foreign key column in child records to its defined default value upon parent deletion.

8 8. Operations on Referencing (Child) Table

8.1 INSERT into Child Table (Violation Risk)

- **Scenario A (Valid Reference):** Insert Course(C4, 'C++', 2).
 - Roll_No = 2 exists in Student.
 - **Result: Allowed.**
- **Scenario B (Invalid Reference):** Insert Course(C5, 'Python', 10).
 - Roll_No = 10 **does not exist** in Student.
 - **Result: Rejected (Referential Integrity Violation).**
- **Scenario C (NULL Insertion):** Insert Course(C6, 'AI', NULL).
 - **Result: Allowed** (if the child foreign key permits NULL).

8.2 DELETE from Child Table (✓ Safe)

- **Scenario:** Executing DELETE FROM Course WHERE Course_ID = 'C1'.
- **Result:** Allowed safely without any violation.
- **Reason:** Parent records do not depend on child records. Removing a child record never leaves the parent table in an invalid state.

8.3 UPDATE in Child Table (Violation Risk)

- **Scenario:** Updating Course row C1's Roll_No from 1 to 99.
- **Result:** Causes a Violation if Roll_No = 99 does not exist in the parent table.

9 9. Master Referential Integrity Exam Matrix

The Golden Rule of Referential Integrity Operations

Parent: Insert Safe, Delete/Update Risk
Child: Delete Safe, Insert/Update Risk

Table 1: Comprehensive Operation Risk Checklist for Exams

Target Table	DML Operation	Status	Key Exam Justification
Parent (Referenced)	INSERT	SAFE	New parent row creates no orphan children.
Parent (Referenced)	DELETE	RISK	May leave child rows referencing a deleted parent.
Parent (Referenced)	UPDATE	RISK	Changing PK value breaks existing child links.
Child (Referencing)	INSERT	RISK	New child FK value must exist in parent table.
Child (Referencing)	DELETE	SAFE	Removing child rows never invalidates parent records.
Child (Referencing)	UPDATE	RISK	Updating FK attribute requires validation against parent.

10 10. SQL Syntax with Referential Actions

```
-- Step 1: Create Parent Table
CREATE TABLE Student (
    Roll_No INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Address VARCHAR(100)
);

-- Step 2: Create Child Table with Cascading Referential Actions
CREATE TABLE Course (
    Course_ID VARCHAR(10) PRIMARY KEY,
    Course_Name VARCHAR(50),
    Roll_No INT,
```

```

CONSTRAINT FK_Course_Student
    FOREIGN KEY (Roll_No) REFERENCES Student(Roll_No)
    ON DELETE CASCADE
    ON UPDATE CASCADE
);

```

11. Multi-Table Cascading Chains

A parent table can be targeted by multiple child tables simultaneously. Furthermore, child tables can themselves act as parent tables for deeper relationships:

Multi-Tier Cascading Architecture

Student (Roll_No) $\xrightarrow{\text{Parent to}}$ Enrollment (Enroll_ID, Roll_No, Course_ID) $\xrightarrow{\text{Parent to}}$ FeePayment (Pay_ID, Enrollment_ID, Amount)

If ON DELETE CASCADE is declared at every level:

Deleting a Student $\implies$ Deletes linked Enrollment rows $\implies$ Deletes linked FeePayment rows

12. Target GATE & Interview Q&A

Q1. Which integrity constraint is maintained by a foreign key?

Referential Integrity.

Q2. Why is inserting into a parent table always safe from referential integrity violations?

Because a parent record exists independently and does not depend on child table entries.

Q3. Why is deleting from a child table always safe?

Removing a child record does not disrupt any parent entity or leave dangling references.

Q4. What happens when a parent row is deleted under default SQL settings?

The operation is rejected (RESTRICT / NO ACTION) if any child rows reference that parent row.

Q5. What is an orphan row?

A child record whose foreign key references a primary key value that no longer exists in the parent table.

Q6. Can ON DELETE SET NULL be applied to a foreign key that is part of a Primary Key?

No. A Primary Key attribute cannot be NULL (violates Entity Integrity).

Q7. Does updating an unreferenced parent key cause a referential integrity violation?

No. If no child rows reference that key value, the update succeeds without violation.

Q8. Can a foreign key reference a candidate key that is not a primary key?

Yes, as long as the referenced column set satisfies uniqueness in the parent table.

13 13. One-Minute Revision & Exam Cheat Sheet

Ultimate Referential Integrity Revision Guide

ENTITY INTEGRITY $\Rightarrow$ Primary Key (Unique + NOT NULL)
REFERENTIAL INTEGRITY $\Rightarrow$ Foreign Key (Valid Parent Reference OR NULL)

Fast Exam Rules for DML Operations:

- Parent INSERT $\rightarrow$ **ALWAYS SAFE**
- Child DELETE $\rightarrow$ **ALWAYS SAFE**
- Parent DELETE / UPDATE $\rightarrow$ **CHECK DEPENDENT CHILDREN**
- Child INSERT / UPDATE $\rightarrow$ **CHECK PARENT EXISTENCE**

DBMS Challenge Progress: Lecture 11 Complete · Next Topic: Super Key, Alternate Key & Composite Key Practice Problems